

Artificial Intelligence 2 KU

Exercise 2

Group 18

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1 Part A: Model-Based Diagnosis

1.1 Representing the Circuit in Boolean Logic

The system description (SD) of a logical circuit represents the intended behavior of each component in the circuit. The correct behavior of a gate i is indicated by the negated abnormality guard $\neg Ab_{Gi}$. For the gates in Figure 1 of the exercise sheet, the SD results in:

$$\begin{aligned} &(\neg Ab_{G_1} \rightarrow (\neg A \wedge B \leftrightarrow G_1)) \wedge \\ &(\neg Ab_{G_2} \rightarrow (A \wedge \neg B \leftrightarrow G_2)) \wedge \\ &(\neg Ab_{G_3} \rightarrow (\neg C \wedge D \leftrightarrow G_3)) \wedge \\ &(\neg Ab_{G_4} \rightarrow (\neg D \wedge C \leftrightarrow G_4)) \wedge \\ &(\neg Ab_{G_5} \rightarrow (G_1 \wedge G_2 \leftrightarrow G_5)) \wedge \\ &(\neg Ab_{G_6} \rightarrow (G_3 \wedge G_4 \leftrightarrow G_6)) \wedge \\ &(\neg Ab_{G_7} \rightarrow (\neg G_5 \wedge \neg G_6 \leftrightarrow G_7)) \end{aligned}$$

1.2 Representing Circuit in DIMACS Format

see content of the "circuit.dimacs" file

The translation of each line in the system description is done according to the following logic, given exemplified for the first line: $(\neg Ab_{G_1} \rightarrow (\neg A \wedge B \leftrightarrow G_1))$

- The bidirectional arrow $X \leftrightarrow Y$ means that X implies Y and Y implies X , so whenever X is true, Y must be true, and whenever Y is true, X must be true:

$$X \leftrightarrow Y \equiv (X \rightarrow Y) \wedge (Y \rightarrow X).$$

- This is logically equivalent to:

$$(\neg X \vee Y) \wedge (\neg Y \vee X)$$

- In the first line of the system description, we have a bidirectional implication on the right and another implication on the left, so the full relationship can be rewritten as:

$$\begin{aligned} &Ab_{G_1} \vee ((\neg A \wedge B) \rightarrow G_1) \wedge (G_1 \rightarrow (\neg A \wedge B)). \\ &(Ab_{G_1} \vee (\neg A \wedge B \rightarrow G_1)) \wedge (Ab_{G_1} \vee (G_1 \rightarrow \neg A \wedge B)). \\ &(Ab_{G_1} \vee A \vee \neg B \vee G_1) \wedge (Ab_{G_1} \vee ((\neg G_1 \vee \neg A) \wedge (\neg G_1 \vee B))) \end{aligned}$$

This final result is a conjunction of 3 clauses, each only consisting out of disjunctions:

$$Ab_{G_1} \vee A \vee \neg B \vee G_1, \quad Ab_{G_1} \vee \neg G_1 \vee \neg A, \quad Ab_{G_1} \vee \neg G_1 \vee B,$$

These are then written in the dimacs file as:

```
12 1 -2 5 0; 12 -5 -1 0; 12 -5 2 0;
```

The output of the dimacs file for the pure system description is given by a SAT solver as:

CPU time: 0.044s

```
SAT -1 -2 -3 -4 -5 -6 -7 -8 -9 -10 -11 12 13 14 15 16 17 18
```

This shows the behavior of the circuit without any observations. All inputs are set to False (-1 to -4), all of the gates' outputs are False (-5 to -11), and all abnormalities are True (12 to 18). The solver can essentially choose any abnormality guards to True to satisfy the structure, so the given output is only one of many possible models, further discussed in the next section.

2 Minimum-Cardinality Diagnosis

How many models are there for the system description without any observations?

A model refers to one complete assignment of truth values to the 18 variables of our system, satisfying all given 21 clauses. In the system description without any observations, there is a large number of possible assignments, that satisfy all conditions.

Using a SAT solver (sharpSAT, [2]), a number of 34992 solutions was found in only 0.019159 seconds.

$\text{:- cell}(R,C), \text{not containWall}(R,C), \text{not lighted}(R,C).$ For the first observation $Obs_1 = G_7 \wedge \neg Ab_{G_1} \wedge \neg Ab_{G_2} \wedge \neg Ab_{G_3} \wedge \neg Ab_{G_4} \wedge \neg Ab_{G_5} \wedge \neg Ab_{G_6} \wedge \neg Ab_{G_7}$, where all gates behave normally and additionally $G_7 = \text{True}$, the system description is expanded with $SD \wedge Obs_1$. In the previously defined circuit.dimacs we only need to add the following lines, where 11 sets $G_7 = \text{True}$ and -12 to -18 sets the abnormality guards to False, and update the number of clauses to 29.

```
11 0; -12 0; -13 0; -14 0; -15 0; -16 0; -17 0; -18 0;
```

By again using the SAT solver, a number of 16 models were found that satisfy the constraints. This is because after enforcing that G_7 is True, and all gates are normal, the only freely choosable parameters left are the 4 inputs A to D. Each of them can take on a binary value, so there are 2^4 possible assignments.

What are the minimum-cardinality diagnoses for (SD,Comps) with Obs_2 ?

The observation $Obs_2 = A \wedge B \wedge \neg C \wedge \neg D \wedge G_7$ the wdimacs file now includes weight for the clauses. For soft clauses, the weight is set to 1, and for hard clauses, it is set to a large number like 1000. All clauses of the system description are defined as hard clauses. The observations are added also as hard clauses with:

1000 1 0; 1000 2 0; 1000 -3 0; 1000 -4 0; 1000 11 0;

The normality of the gates is now only defined with soft clauses:

1 -12 0; 1 -13 0; 1 -14 0; 1 -15 0; 1 -16 0; 1 -17 0; 1 -18 0;

In total this leads to 33 clauses. The purpose of a MaxSAT solver is now to not only find the number of possible assignments to fully satisfy all the clauses. Instead, it finds a solution that satisfies all hard clauses and minimizes the violated soft clauses. Using a MaxSAT solver (open-wbo, [1]), the minimum-cardinality diagnosis found is an empty set, i.e. none of the soft clauses need to be violated to also fulfill the hard clauses. More concretely, no gates need to be faulty to make observation 2, so the output is the same as our system definition:

1 2 -3 -4 -5 -6 -7 -8 9 10 11 -12 -13 -14 -15 -16 -17 -18

What are the minimum-cardinality diagnoses for (SD,Comps) with Obs_3 ?

For observation $Obs_3 = A \wedge B \wedge \neg C \wedge \neg D \wedge \neg G_7 \wedge Ab_{G_7}$, the only changes are that G_7 is normal and set to True, both defined as hard clauses.

Using the same MaxSAT solver, the found optimal solution has a cost of 2, so at least two of the soft clauses need to be violated in order to fulfill the hard clauses. This means two of the gates need to be faulty to lead to observation 3.

The optimal model given by the solver shows, that G_5 and G_6 are faulty:

1 2 -3 -4 -5 -6 -7 -8 9 10 -11 -12 -13 -14 -15 16 17 -18

References

- [1] Ruben Martins, Vasco Manquinho, and Ines Lynce. Open-wbo: A modular maxsat solver. In *International Conference on Theory and Applications of Satisfiability Testing*, pages 438–445. Springer, 2014.
- [2] Marc Thurley. sharpsat: A sat model counter. <https://github.com/marcthurley/sharpSAT>. Accessed: 2026-05-04.